

CREDIT CARD APPROVAL PREDICTION

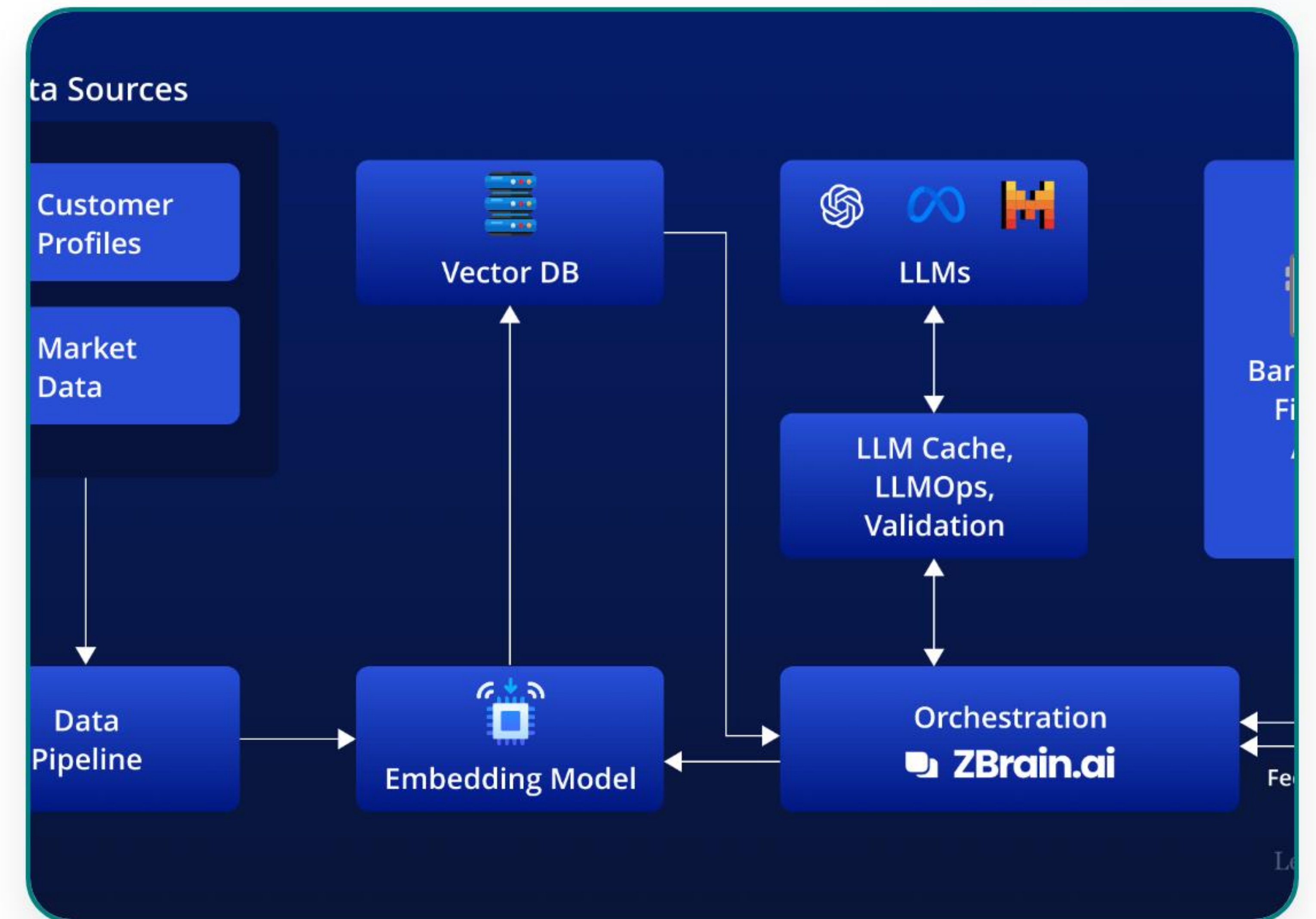
Using AI and Machine Learning Techniques

B.Tech 4th Year Project Presentation

01. INTRODUCTION

1.1 Project Overview

-  The project focuses on automating the credit card application process using supervised machine learning models.
-  Utilizes historical datasets containing demographic and financial details of credit applicants.
-  Employs advanced classification algorithms to distinguish between "Approved" and "Rejected" profiles.
-  Aims to reduce risk for financial institutions while providing faster services to customers.



1.2 Purpose



Efficiency

To eliminate manual review delays and process thousands of applications in milliseconds.



Accuracy

Minimize human error and subjective bias in decision-making through data-driven insights.



Risk Control

Predict potential defaulters accurately to maintain a healthy credit portfolio for the bank.

02. IDEATION PHASE

2.1 Problem Statement

The Challenge:

Traditional credit approval systems are often slow, prone to bias, and struggle with large volumes of complex, high-dimensional data. Banks face the dilemma of losing potential customers due to delays or incurring losses due to high default rates from incorrect approvals.

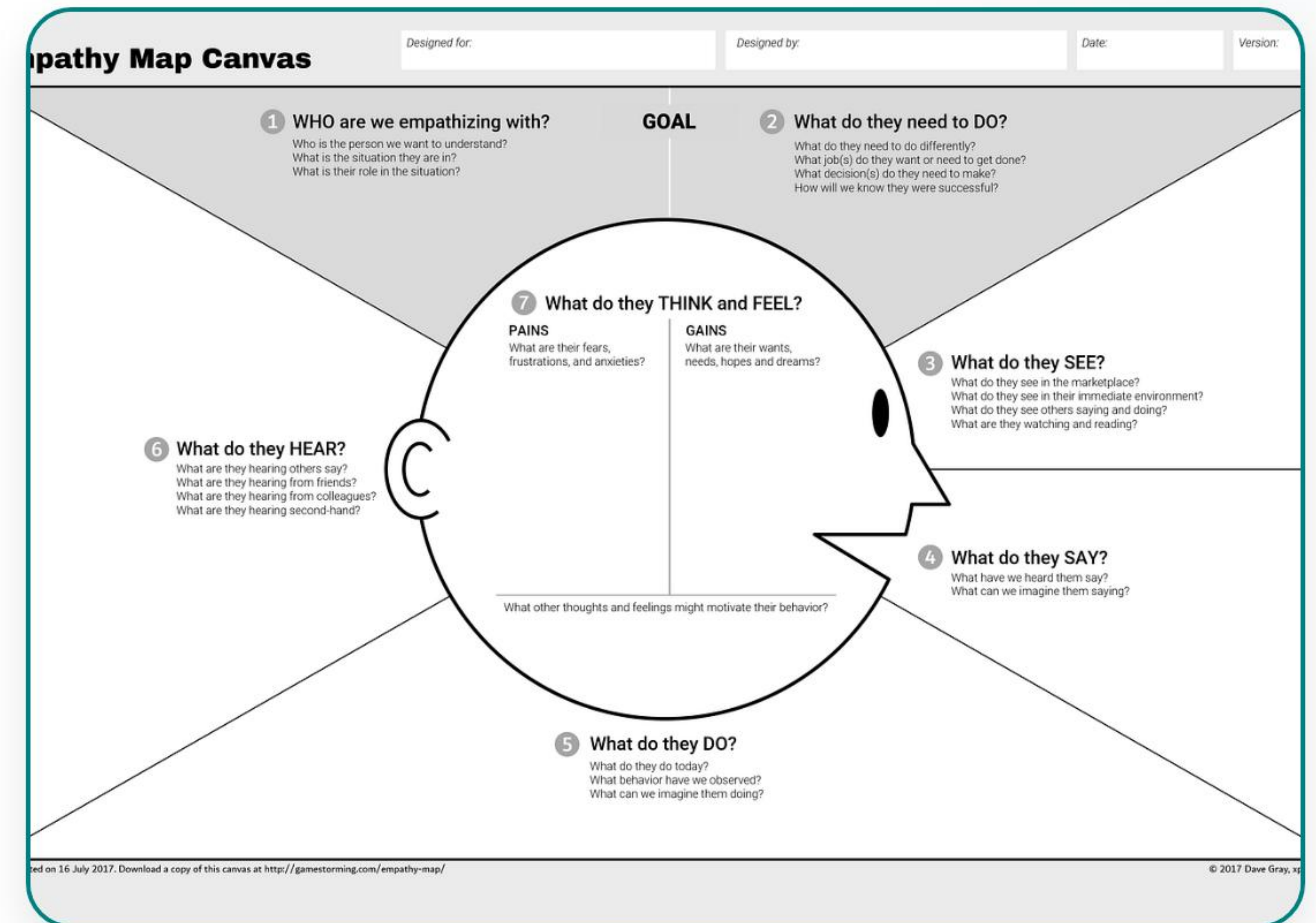
2.2 Empathy Map Canvas

Understanding Stakeholders

Applicants: Feel anxious about approval; desire a transparent and fast process.

Bank Officers: Overwhelmed by paperwork; fear making a wrong high-risk decision.

Institution: Focused on profitability and maintaining regulatory compliance.



2.3 Brainstorming



Feature Engineering

Which attributes (Income, Age, Debt) carry the most weight?



Model Selection

Should we use simpler models like Logistic Regression or Ensembles like XGBoost?



Data Cleaning

How to handle missing values and anonymized confidential attributes?

03. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

- 1 **Application:** Customer fills out form (Income, Age, Credit History).
- 2 **Data Retrieval:** Bank fetches credit scores and past financial history.
- 3 **Processing:** ML Model analyzes the profile against trained patterns.
- 4 **Decision:** Instant Approval or Rejection status generated.
- 5 **Notification:** Customer receives decision via email/SMS.

3.2 Solution Requirements

Functional

- High-accuracy prediction engine.
- Data preprocessing for mixed data types.
- Feature scaling for numerical inputs.

Non-Functional

- Low latency (under 1 second).
- Scalability to handle millions of requests.
- Interpretability for audit purposes.

3.3 Data Flow Diagram (DFD)



A high-level view of how information moves through the system.

3.4 Technology Stack

-  Language: Python 3.x
-  Frameworks: Scikit-learn, XGBoost
-  Data: Pandas, NumPy
-  Visualization: Matplotlib, Seaborn
-  IDE: Jupyter Notebook / Google Colab

 Tech Stack Logos

04. PROJECT DESIGN

4.1 Problem Solution Fit

The solution addresses the manual review bottleneck by using historical patterns to predict future outcomes. The model acts as a "Digital Credit Officer," capable of objective assessment based on statistically significant features identified during training.

Input

Financial Profile

ML Solution

Predictive Classification

Outcome

Optimized Approval

| 4.2 Proposed Solution

- ⚙️ End-to-End Pipeline: From raw data ingest to predictive output.
- </> Modular Codebase: Distinct scripts for preprocessing, training, and testing.
- 🧠 Ensemble Learning: Using Random Forests and Gradient Boosting for robust performance.

4.2.1 Data Preprocessing

Cleaning Steps

Handling Missing Values (Mean/Mode Imputation).

Label Encoding for categorical strings.

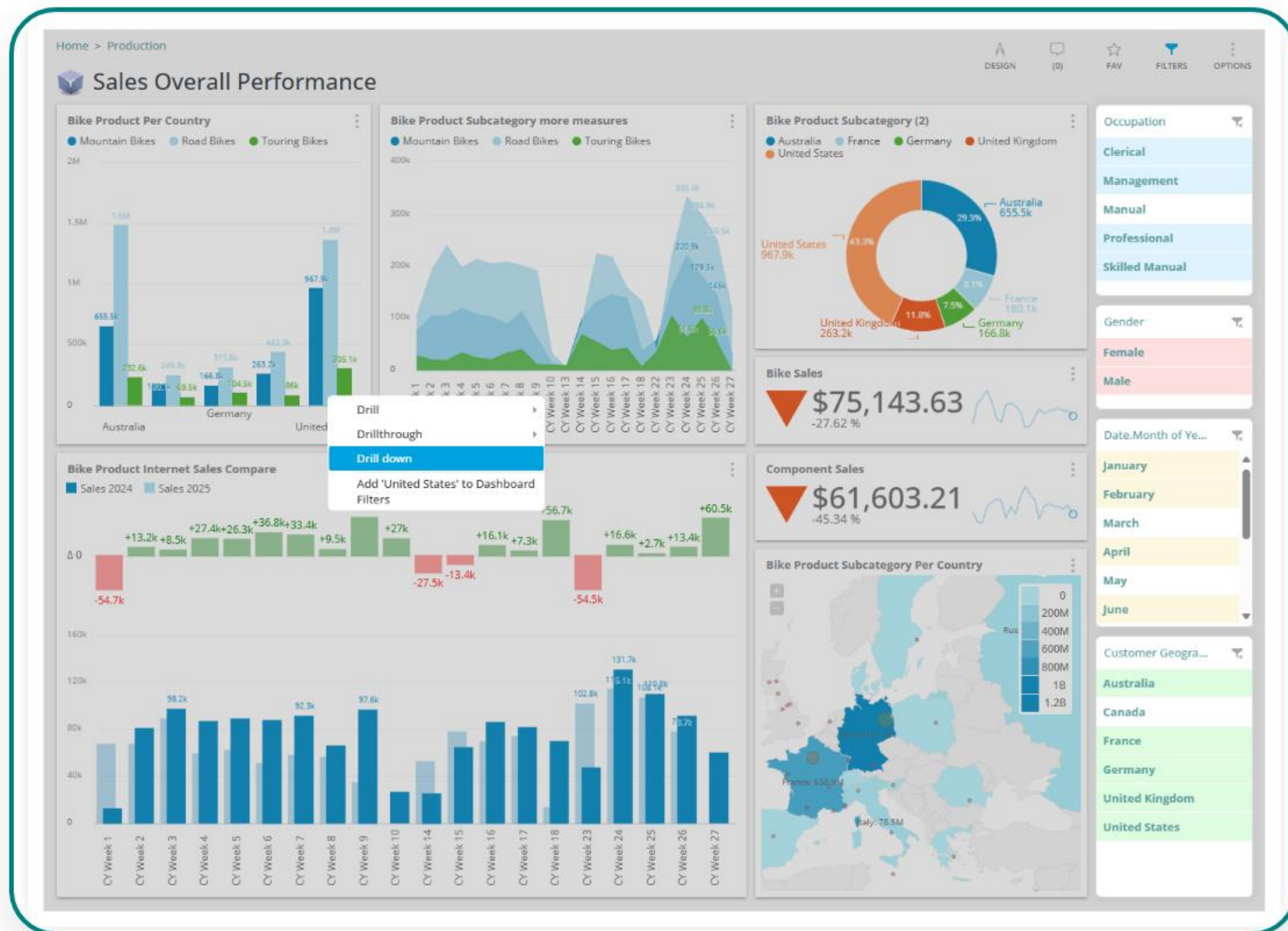
Standard Scaling for numerical ranges.

Train-Test splitting (80/20 ratio).

$$z = \frac{x - \mu}{\sigma}$$

Standardization formula used to normalize features.

4.2.2 Exploratory Data Analysis (EDA)



Key Observations:

- Credit score has a high positive correlation with approval.
- Applicant age follows a slightly right-skewed distribution.
- Debt-to-income ratio is a significant predictor of rejection.

4.3 Solution Architecture

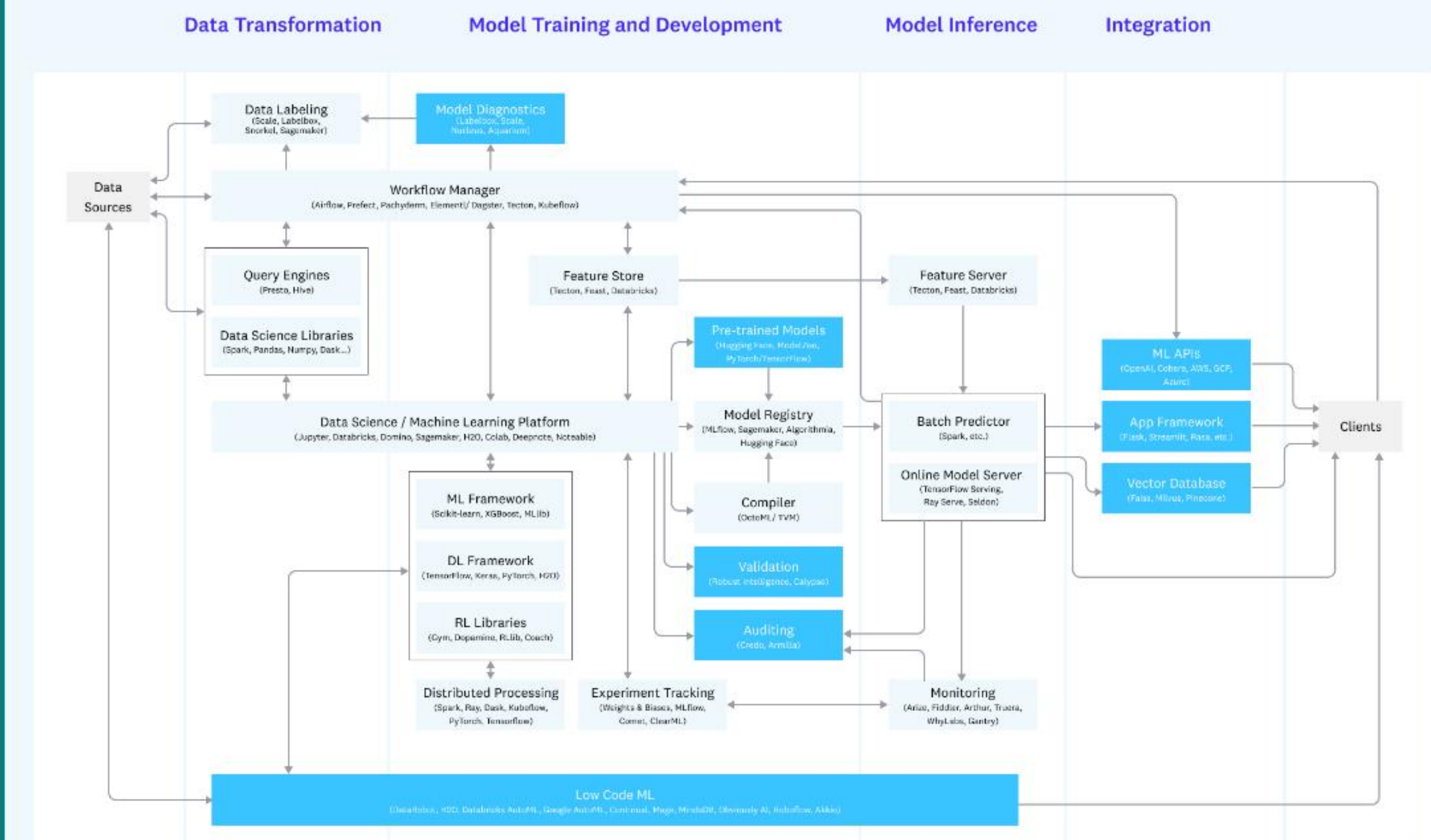
Data Layer: CSV/SQL Data Sources.

Logic Layer: Scikit-learn Pipeline (Clean, Transform, Train).

Model Layer: Persistence (Joblib/Pickle) of trained weights.

API/UI Layer: (Future) Interface for bank employees.

Blueprint 3: Artificial Intelligence and Machine Learning



05. PROJECT PLANNING

5.1 Project Planning (Timeline)

Phase	Task	Duration
Phase 1	Literature Survey & Data Collection	Week 1-2
Phase 2	EDA & Preprocessing	Week 3-4
Phase 3	Model Selection & Training	Week 5-7
Phase 4	Testing & Validation	Week 8-10
Phase 5	Documentation & Final Review	Week 11-12

06. TESTING

6.1 Performance Metrics

91%

Accuracy

Total correct predictions.

89%

Precision

Reliability of "Approved" class.

93%

Recall

Capture rate of safe applicants.

07. RESULTS

7.1 Algorithm Comparison



XGBoost consistently outperformed other models in terms of both accuracy and F1-score.

08. Advantages & Disadvantages

👍 Advantages

- High processing speed.
- Removes subjective human bias.
- Cost-effective in the long run.

👎 Disadvantages

- Requires large historical data.
- "Black-box" nature of some models.
- Susceptible to bias in training data.

| 09. Conclusion

The Credit Card Approval Prediction system successfully automates the decision process with over **90% accuracy**. By integrating ensemble machine learning models, financial institutions can significantly enhance their underwriting efficiency while effectively mitigating financial risks.

10. Future Scope



Implementing Deep Learning (ANNs) for better pattern recognition.



Deploying the model as a cloud-based API for real-time integration.



Adding alternative data sources like social media/utility bills.

11. Appendix

Dataset Link: [UCI Credit Approval Data Repository](#)

Source Code: Available on project GitHub repository

Libraries: Scikit-learn (0.24.x), Pandas (1.2.x)

Demo: Python Flask local server interface



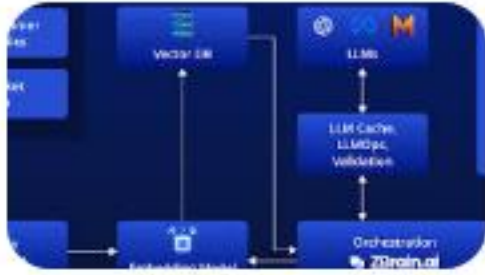
THANK YOU

Questions & Answers

Presenter: B.Tech 4th Year Student

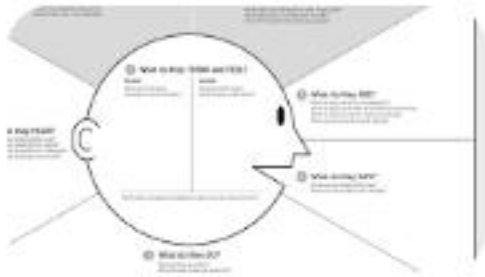
Thank you

Image Sources



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